THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 6 — SCIENCE · HARD LEVEL CHAPTER TEST SET 3

Chapter 6: **Materials Around Us** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Why is jute used for making sacks? 1

(a) it is transparent **(b)** it is brittle
(c) it is soluble **(d)** it is strong and cheap

2. A fabric that does not absorb water is best for making: 1

(a) raincoats **(b)** handkerchiefs **(c)** bandages **(d)** towels

3. Why is aluminium foil used to wrap food? 1

(a) it can be rolled into thin sheets and is non-toxic
(b) it is transparent
(c) it dissolves
(d) it is magnetic

4. A material that is rough, hard and absorbs water could be: 1

(a) plastic **(b)** glass **(c)** steel **(d)** a brick

5. A fabric that absorbs sweat is best for: 1

(a) tents **(b)** umbrellas **(c)** raincoats **(d)** innerwear

6. Which property is common to a mirror and a polished steel plate? 1

(a) solubility **(b)** lustre **(c)** floating **(d)** transparency

7. Which pair of properties best suit a good insulating handle? 1

(a) transparent and soluble
(b) shiny and magnetic
(c) good conductor of heat
(d) poor heat conductor and non-conductor of electricity

8. Which of the following is an example of using a property of material wisely? 1

(a) using wood for a stove **(b)** using chalk for tyres
(c) using glass for a greenhouse **(d)** using paper for pipes

9. Which of these changes shows a property that depends on temperature? 1

(a) copper is shiny **(b)** wood is hard
(c) sugar dissolves faster in hot water **(d)** glass is transparent

10. Which factor must be kept the same when comparing how fast sugar and salt dissolve? 1

(a) name **(b)** smell
(c) colour **(d)** temperature and amount of water

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Why do we call cork a 'light' material? 2

12. Name the property used to separate iron filings from sand and the property used to separate sand from salt. 2

13. Why can't we make a pot from sugar? 2

14. Why are dark-coloured utensils sometimes used in cooking? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. How would you test whether a cloth is cotton or nylon without using fire? 3

16. Why must the same amount of water and heat be used when comparing rods for conduction? 3

17. Explain why transparent materials may still cast faint shadows. 3

OR

Explain why materials like diamond and graphite, both made of carbon, differ in hardness and use. 3

18. Explain with examples why lustre alone cannot identify a metal. 3

OR

Why are objects painted? Link to properties of materials. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Explain how knowledge of materials has helped in the progress of human beings: stone age, bronze age, iron age and modern materials. 5

OR

Describe how to choose the best materials for a school bag. Give properties needed and examples. 5

20. A shopkeeper says 'Gold is hard like diamond'. Test the claim with properties. 5

OR

Discuss how reuse and recycling of materials such as paper, glass, metals and plastic help the environment. 5

END OF QUESTION PAPER — Total: 40 Marks

MASTERMIND